
Paper vs. Ours, with Custodial:

A Constraint-by-Constraint Comparison

How custodial symmetry changes (and does not change)
each RS flavor / electroweak constraint we run.

June 2026

1 Introduction

This note answers one question: **how does custodial symmetry** $(\text{SU}(2)_L \times \text{SU}(2)_R \times \text{U}(1)_X \times P_{LR})$ **affect the constraints we actually run in our minimal-RS scan?** For each observable we put the **published figure on the left** and **our fixed-code reproduction on the right**, focusing on what custodialization does to the floor.

The one-line answer. Custodial symmetry is an *electroweak* cure, not a *flavor* cure. It protects the oblique T parameter and the $Z b_L \bar{b}_L$ vertex, pulling the EW floors down from $\sim 16\text{--}20$ TeV (ours, minimal) to a few TeV. It does **nothing** to the ε_K wall, which stays at tens of TeV — because the binding kaon operator is the color-octet KK-gluon LR/C_4 operator, an electroweak singlet that the custodial group cannot touch. This is not a feature of our proxy: Blanke et al. (0809.1073) computed the *full* $\Delta F = 2$ *inside the custodial model* and still found $M_{KK} \gtrsim 18\text{--}30$ TeV from ε_K .

Lane note. The ε_K comparisons here use our **anarchic baseline (lane A)** — the literature strawman — so they line up against Bauer / CFW / Blanke. Production as run (lane B, fit-aligned MFV) is a sharper wall at ≈ 7 TeV, and the FPR/aligned ideal (lane C) is ≈ 2 TeV. Custodial does not move *any* of these flavor lanes; only *flavor alignment* does. Canonical lane definitions: docs/MODEL_CONVENTIONS.md.

2 Real physics vs. proxy limitation

Before the comparisons, one honest caveat. When our code shows custodial touching **only** S, T, U and $Z \rightarrow b\bar{b}$ and *nothing* in flavor, that is $\sim 90\%$ **real physics** and $\sim 10\%$ **a proxy limitation**. The two must not be conflated.

REAL (literature-confirmed) vs. PROXY (genuinely missing)

REAL — the ε_K wall is genuinely custodial-blind. ε_K is dominated by the **KK-gluon** LR/ C_4 operator. The KK gluon is a *color octet, electroweak singlet* — uncharged under the custodial group, which is purely electroweak — so custodialization *cannot* change the leading ε_K amplitude. This is confirmed by a **full calculation, not our proxy**: **Blanke 2008** (0809.1073) computed the complete $\Delta F = 2$ inside the custodial model (custodians, EW KK bosons, everything) and still found $M_{KK} \gtrsim 18\text{--}30$ TeV. Bauer II (0912.1625): C_4 “is a pure QCD effect ... insensitive to the precise embedding of the electroweak gauge symmetry.”

PROXY LIMITATION — what we genuinely omit. Our implementation is a **leading-order tree EW proxy**, not the full custodial model. It does NOT implement: the *custodian KK-fermion spectrum* (charge-5/3 states, bidoublet partners); the *subleading custodial flavor effects* custodial *does* touch (EW KK bosons — KK- Z , Z' , Z_H — contributing to $\Delta F = 2$; custodian loops in $b \rightarrow s\gamma$); the *full* ΔT (singlet-only, dropping the dominant negative bidoublet ΔT); a non-trivial b_R representation (g_R^b is hard-zeroed).

Therefore: the headline “*custodial fixes EW ($\sim 2\text{--}3$ TeV) but not the ε_K wall (tens of TeV)*” is REAL and literature-confirmed. The stronger-sounding “*custodial does literally nothing to any flavor observable*” — what our code prints — is partly a proxy limitation. **The plots below use the LEADING custodial physics only, which we implement correctly.** This proxy is adequate for the RS-vs-SM note’s thesis; it is *not* adequate for custodial phenomenology (A_{FB}^b , $b \rightarrow s\gamma$, the subleading custodial $\Delta F = 2$, the precise custodial EW floor) — that is the deferred full custodial build (roadmap D), scoped by the four proxy items in §5.

3 Side-by-side: paper vs. ours, constraint by constraint

In every composite below the **PAPER** panel (the cropped published figure) is on the left and **OURS** (our fixed-code reproduction) is on the right.

3.1 (a) S, T, U oblique — custodial protects T , not S

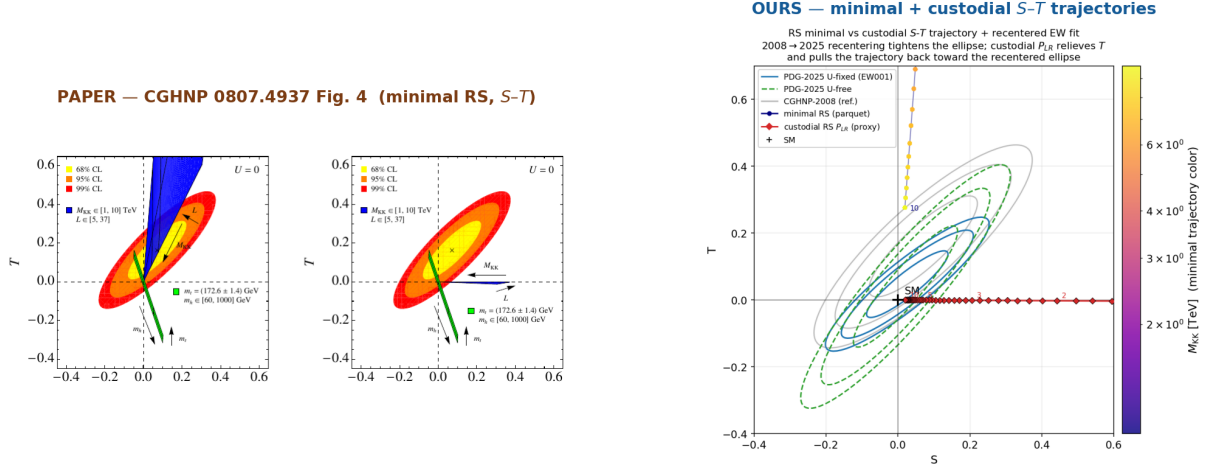


Figure 1: Oblique S - T . *Left (paper):* CGHNP 0807.4937 Fig. 4, minimal RS sweeping a wedge that climbs steeply in T (the volume-enhanced “RS T problem,” $T \propto \pi L / (2 \cos^2 \theta_W) \cdot v^2 / M_{KK}^2$). *Right (ours):* the same minimal trajectory (it shoots up in T) plus the **custodial P_{LR} trajectory**, which runs almost flat along the S -axis ($T \rightarrow 0$) — the ADMS mechanism ($SU(2)_R$ kills the $\log(L \approx 30)$ enhancement, $\Delta T \propto +L \rightarrow -1/L$). Custodial relieves the irreducible oblique floor: **our minimal ~ 16 – 20 TeV $\rightarrow$ custodial ~ 6 TeV**; the residual is set by the *unprotected* S (literature custodial floor ~ 2.5 – 3 TeV, S -limited). S carries no symmetry protection, so it sets the wall.

3.2 (b) $Z \rightarrow b\bar{b} - P_{LR}$ protects g_L^b at tree, loops re-introduce it

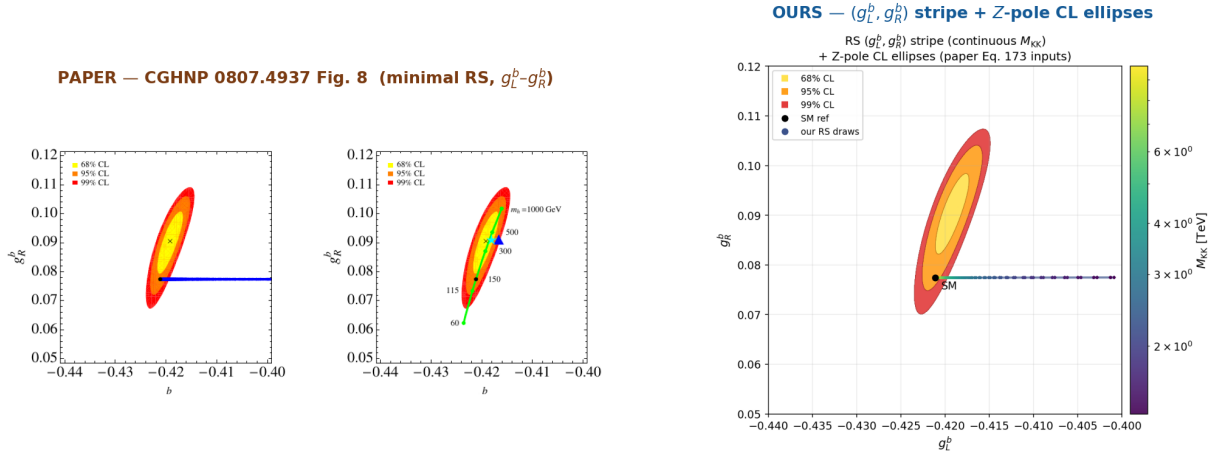


Figure 2: $Z \rightarrow b\bar{b}$ couplings, minimal. *Left (paper):* CGHNP 0807.4937 Fig. 8, the (g_L^b, g_R^b) plane with the Z-pole 68/95/99% CL ellipses and the minimal-RS scatter. *Right (ours):* our (g_L^b, g_R^b) stripe overlays the same ellipses and SM point (sanity check). With the B1 fix the total coupling shift is gauge-KK dominated and moves g_L^b to less-negative values as M_{KK} drops — the CGHNP direction. In *minimal* RS this bites only to ~ 5 TeV (the old 25–30 TeV “ $Z \rightarrow b\bar{b}$ floor” was a sign-bug artifact).

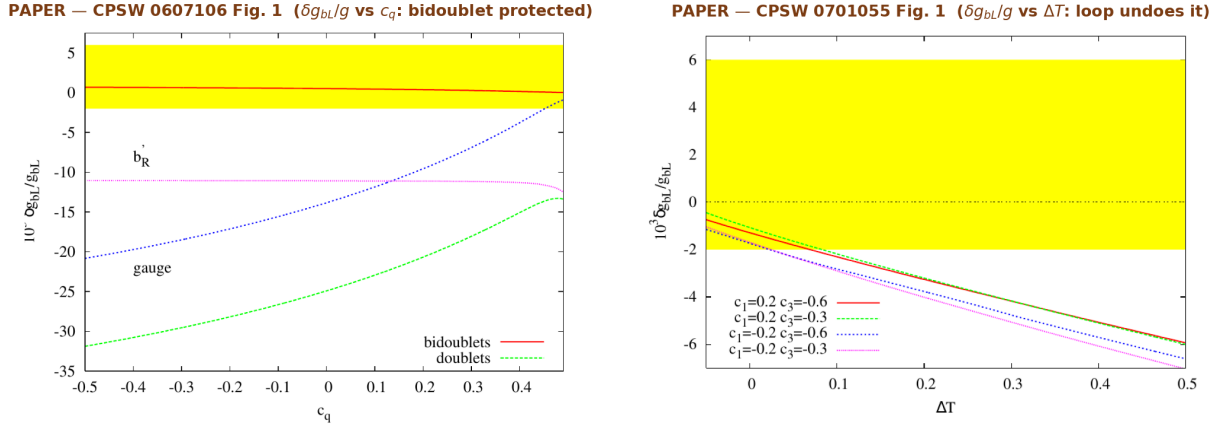
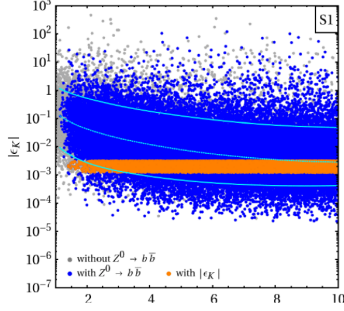


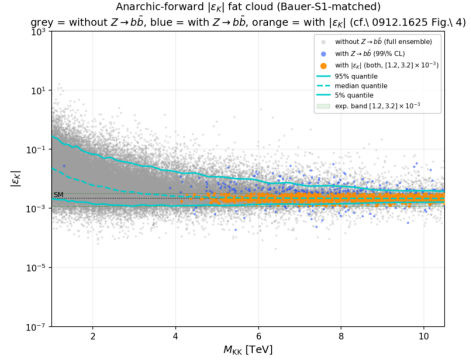
Figure 3: $Z \rightarrow b\bar{b}$ under custodial (CPSW). *Left:* CPSW hep-ph/0607106 Fig. 1 — $\delta g_{bL}/g$ vs. c_q . The **bidoublet** embedding (red) sits inside the yellow allowed band: P_{LR} with $b_L \in (2, 2)_{2/3}$ zeroes δg_L^b at tree level (ACDP), whereas the unprotected “gauge”/“doublets” curves run far below. *Right:* CPSW hep-ph/0701055 Fig. 1 — $\delta g_{bL}/g$ vs. ΔT . The top-partner **one-loop** corrections **re-introduce** a negative δg_L^b , correlated with $+\Delta T$; the curves exit the 2σ band by $\Delta T \approx 0.2\text{--}0.5$. So P_{LR} (ACDP) zeroes δg_L^b at tree, but the top partners (CPSW) bring it back at loop — the custodial $Z \rightarrow b\bar{b}$ story is a tree-protection that loops partially undo, which our loop proxy reproduces (CPSW Eqs. 13–15).

3.3 (c) ε_K — the wall does NOT move under custodial

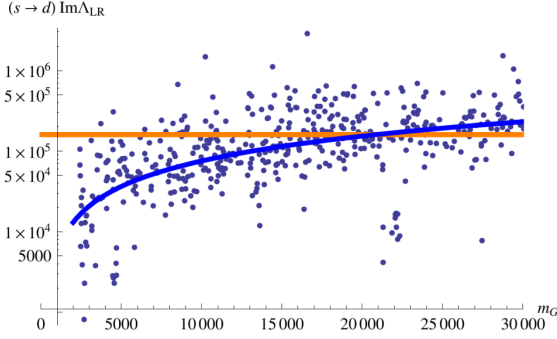
PAPER — Bauer 0912.1625 Fig. 4 (S1) (minimal anarchic ε_K)



OURS — Lane-A anarchic $|\varepsilon_K|$ cloud



PAPER — CFW 0804.1954 Fig. 1 (minimal anarchic $\text{Im}\Lambda_{LR}$ vs m_G)



OURS — ε_K wall (minimal + Blanke custodial + ours)

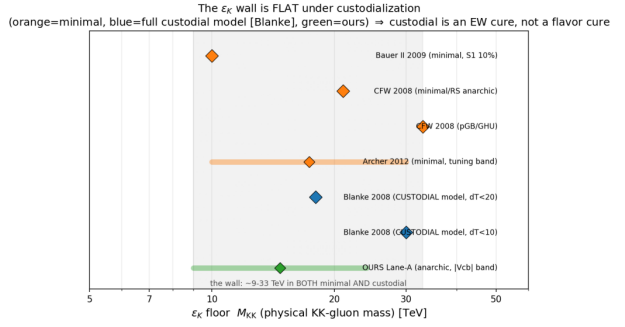


Figure 4: ε_K — the MINIMAL anarchic wall. *Top-left (paper):* Bauer 0912.1625 Fig. 4 (S1), the anarchic $|\varepsilon_K|$ cloud spanning ~ 6 decades; the 95%-quantile floor sits at $M_g^{(1)} \gtrsim 10$ TeV. *Top-right (ours):* our Lane-A anarchic $|\varepsilon_K|$ cloud (Bauer-S1-matched), same ~ 6 -decade spread, same grey/blue/orange colouring and 5/50/95% quantiles. *Bottom-left (paper):* CFW 0804.1954 Fig. 1, the anarchic $\text{Im}\Lambda_{LR}$ ($s \rightarrow d$) KK-gluon scan vs. m_G , crossing the bound (orange) around $m_G \approx 20$ TeV. *Bottom-right (ours):* our ε_K floor scoreboard — minimal literature anchors, the Blanke custodial anchors, and our Lane-A band, all in the ~ 9 –33 TeV wall. The point: in *minimal* RS the wall is tens of TeV.

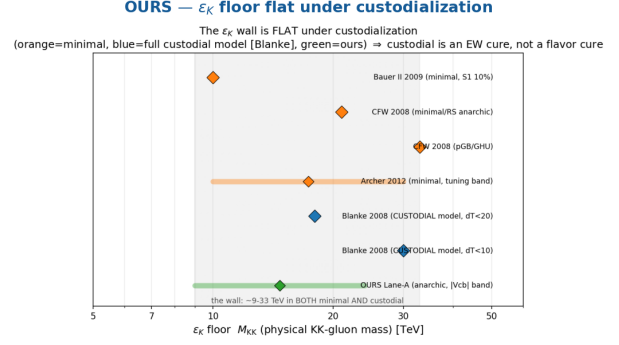
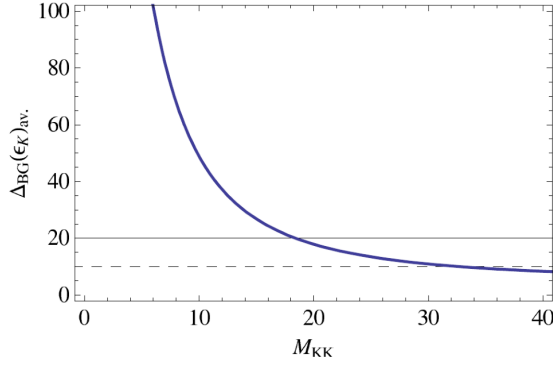


Figure 5: ε_K — the CUSTODIAL model gives the SAME wall. *Left (paper):* Blanke 0809.1073 Fig. 9 — the ε_K fine-tuning $\Delta_{BG}(\varepsilon_K)$ vs. M_{KK} computed **inside the full custodial model**. The curve crosses $\Delta < 20$ at $M_{KK} \approx 18$ TeV and $\Delta < 10$ at $M_{KK} \approx 30$ TeV — the wall **survives custodialization**. *Right (ours):* our flat-wall scoreboard puts the Blanke *custodial* anchors (18/30 TeV) right on top of the *minimal* Bauer/CFW/Archer anchors and our Lane-A band. **The ε_K wall does not move under custodial** — in the literature (Blanke’s full calculation) and in ours. This is the crux: custodial is an EW cure, not a flavor cure.

3.4 (d) Synthesis: the custodial decoupling ladder

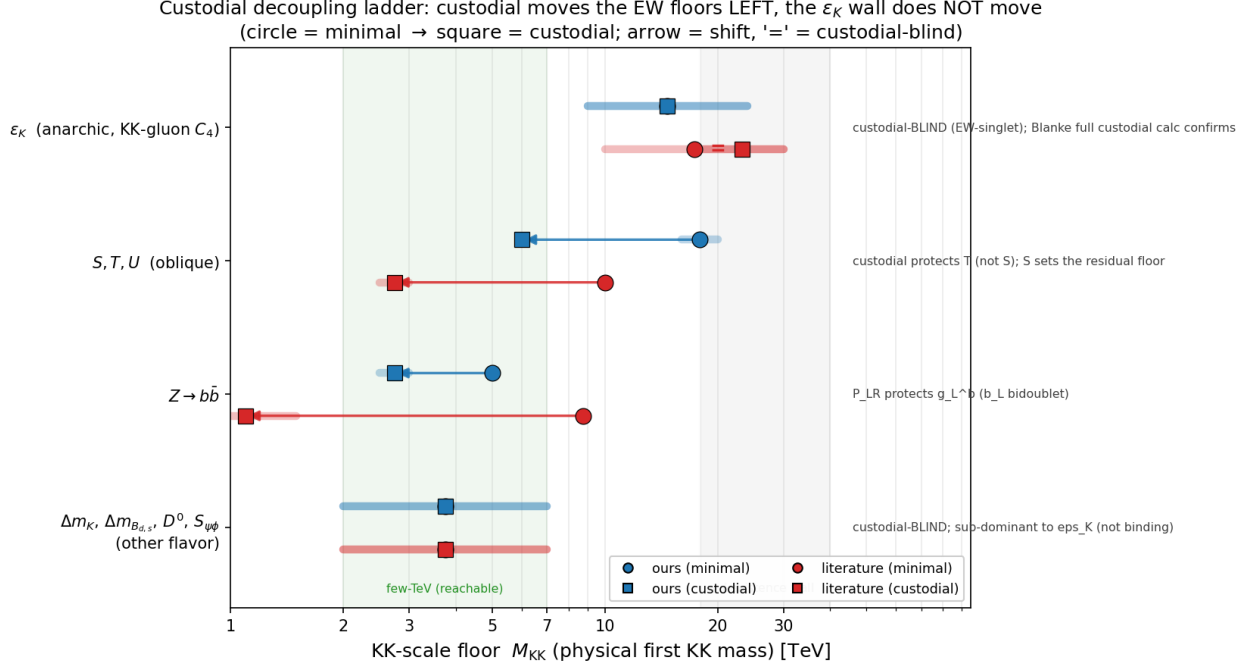


Figure 6: The whole story on one axis. Per-constraint KK-mass floor, minimal (circle) $\rightarrow$ custodial (square), ours (blue) and literature (red). The **electroweak** floors (S, T, U and $Z \rightarrow b\bar{b}$) move **left** under custodial (arrows); the ε_K wall and the other flavor constraints **do not move** (“=”). Custodial brings the EW sector into the few-TeV reachable band (green) but leaves the ε_K /existence wall at tens of TeV (grey).

4 Provenance, attribution, and literature floors

These tables condense the three custodial provenance documents in docs/: CUSTODIAL_PROVENANCE.md (where every equation comes from), CUSTODIAL_PLOTS_AND_ATTRIBUTION.md (who computed what), and CUSTODIAL_LITERATURE_CATALOG.md (per-paper floors). Status tags: **S** = sourced (exact published equation), **T** = faithful truncation (documented simplification), **P** = proxy (correct parametric form, underived coefficients/inputs).

4.1 Master provenance: where every custodial equation comes from

Code quantity	Form	Source	Tag	Gap?
minimal ΔT	$x_1^2 \pi L / (2c_W^2)(v/M)^2$	CGHNP (147)	T	leading- L (0.04% cons.)
custodial ΔT	$-x_1^2 \pi / (4c_W^2 L)(v/M)^2$	CGHNP (153) $\leftarrow$ ADMS	S	in-code cite missing
ΔS	$c_S(v/M)^2$, $c_S=30$	PDG-2025	S	—
ΔU	0	CGHNP/ADMS	S	—
minimal g_L^b	$+(m_b^2/2M^2)B_d$	CGHNP (170)	S	convention caveat
minimal g_R^b	$-(m_b^2/2M^2)B_Q$	CGHNP (170)	S	convention caveat
custodial $g_L^b \rightarrow 0$	P_{LR} zeroing	ACDP (3)–(12)	S	—
custodial g_L^b residual	$\kappa_b(1/L)g_L^{b,\min}$	—	P	ad hoc knob
custodial $g_R^b=0$	$g_R^b \rightarrow 0$	ACDP (22)/Tab. 1	T	rep not pinned
loop δg_L^b singlet	CPSW (14)	CPSW-ob (14)	S	—
loop δg_L^b bidoublet	CPSW (15)	CPSW-ob (15)	S	—
loop ΔT singlet	CPSW (13)+(35)	CPSW (13)/(43)	T	drops neg. bidoublet ΔT
loop $\delta g_R^b=0$	0	CPSW-ob (p.10)	S	—
loop inputs ξ, ρ	$M_t=\rho_t M, \dots$	—	P	breaks wavefn. corr.

Bottom line: *every closed-form equation* in our custodial implementation is faithfully reproduced from a local peer-reviewed source PDF. The genuine freedom lives in **four proxy choices, not in the equations** (§5).

4.2 Attribution: who derived the mechanism, who computed the form we use

Process	What custodial does	Derived the mechanism	Computed the form we use
Oblique T	$SU(2)_R$ kills the $\log L$ T -enhancement; $\Delta T \propto +L \rightarrow -1/L$	ADMS 2003 (hep-ph/0308036)	CGHNP 2008 (0807.4937) Eq. 153
Oblique S	NOT protected — sets the residual floor	(no protection)	PDG-2025 ($c_S=30$)
$Z \rightarrow b\bar{b} g_L^b$	P_{LR} zeroes δg_L^b at tree, $b_L \in (2, 2)_{2/3}$	ACDP 2006 (hep-ph/0605341)	ACDP (3)–(12); CGHNP Eq. 170
$Z \rightarrow b\bar{b} g_R^b$	$b_R \in (1, 1)_{-1/3} \Rightarrow \delta g_R^b=0$ (P_C)	ACDP 2006 Table 1	ACDP Table 1 row 2
$Z \rightarrow b\bar{b}$ loop	top-partners re-introduce $-\delta g_L^b, \Delta T$	CPSW 2006 (hep-ph/0607106)	CPSW 2007 (0701055) Eqs. 13–15, 35
$\varepsilon_K, \Delta m_K$	NOTHING (KK-gluon C_4 , EW-singlet)	CFW 2008; Bauer II 2009; custodial: Blanke 2008	GGMS hadronic ME in <code>deltaf2.py</code>
$\Delta m_{B_d}, \Delta m_{B_s}$	nothing (leading); partial EW-KK subleading	Bauer II 2009; Blanke 2008	same <code>deltaf2.py</code> path
D^0 mixing	nothing (up-sector KK-gluon)	GGNP 2009 (funnel)	our D -funnel post-proc.

In words. ADMS first showed $SU(2)_R$ protects T ; ACDP first showed P_{LR} protects $Zb_L\bar{b}_L$ and tabulated the b_R rep $\rightarrow \delta g_R^b$ map; CPSW computed the top-partner one-loop ($-T$ bidoublet / $+T$ singlet, and the loop δg_L^b); CGHNP distilled the custodial T into the compact Eq. 153 our code uses; Blanke computed the *full* $\Delta F = 2$ in the custodial model and showed ε_K stays $\gtrsim 18$ –30 TeV; CFW and Bauer II computed the anarchic ε_K wall we reproduce as Lane A. We wired these published forms into the code, with four proxy choices that are ours, not theirs.

4.3 Per-paper floors: custodial EW vs. ε_K

Paper	Custodial EW floor	ε_K floor	Key statement
ADMS 2003 (0308036)	$M_{KK} \sim 3$ TeV ($S+Zb_L$)	not addressed	origin of custodial-for- T ; ε_K a separate axis
CPSW 2006/07	$M_g^{(1)} \sim 2.5$ –3 TeV (S -set)	absent	custodial lets KK be light; tree Zb_L protected, loop partly undoes it
CGHNP I 2008 (0807.4937)	$M_{KK} > 2.4$ TeV (~ 6 TeV M_1)	~ 10 –20 TeV M_1	“four cures” are cures for T , not ε_K
CFW 2008 (0804.1954)	~ 3 TeV (S)	21 TeV (RS) / 33 TeV (GHU)	most explicit EW-vs-flavor split
Blanke 2008 (0809.1073)	2–3 TeV (EW prec.)	18 TeV ($\Delta < 20$) / 30 TeV ($\Delta < 10$)	THE custodial-model $\Delta F=2$ paper — ε_K wall survives custodial
Bauer II 2009 (0912.1625)	Zb_L removable	$M_g^{(1)} \gtrsim 10$ TeV (S1) $\rightarrow$ S2 align ~ 3 TeV	C_4 “a pure QCD effect . . . insensitive to the EW embedding”
Archer 2012 (1201.1561)	$M_{KK} \sim 1.5$ –2 TeV	$M_{KK} \sim$ 10/22/30 TeV (tuning)	custodial does not protect S or gauge-fermion couplings

Ours, lane-tagged, vs. the literature: S, T, U existence floor (non-custodial) **18–20 TeV** $\rightarrow$

literature custodial drops to $\sim 2\text{--}3$ TeV (what a full custodial build would buy); $Z \rightarrow b\bar{b}$ floor (post-B1) ~ 5 **TeV** (P_{LR} -protected Zb_L); ε_K Lane A $\sim 9\text{--}24$ **TeV** (matches Blanke 18–30, CFW 21–33, Bauer II ~ 10 , Archer 10–30 — *and custodial does not move it*); ε_K Lane C (aligned) ~ 2 **TeV** (alignment, not custodial).

5 The four proxy gaps — and what a full custodial build would add

These are the *only* places our custodial implementation is not a faithful copy of a published equation. Each is the target of the deferred full custodial build (roadmap D).

The punch-list (CUSTODIAL_PROVENANCE.md §3)

1. **Custodial g_L^b residual $\kappa_b(1/L)g_L^{b,\min}$ — UNSOURCED.** Neither ACDP nor CGHNP writes a $1/L$ -suppressed P_{LR} -breaking residual; κ_b is a free $O(1)$ dial. *To source:* derive the P_{LR} -breaking operator coefficient in the production rep $((2,2)_{2/3}$ for q_L) and show it scales as $1/L$. *Until then:* a phenomenological knob, not a derived residual.
2. **Top-partner loop inputs $\xi_s, \xi_q, \xi_\chi, \rho_t, \rho_q, \rho_\chi$ — PROXY.** The formulas (13–15) are exact, but the six KK/mixing masses are re-parametrized as independent $O(1)$ dials. CPSW’s physical claim is that these masses are *locked together* by the 5D wavefunctions; the repo breaks that correlation. *To source:* back-derive the six masses from the 5D overlap integrals (CPSW App. A) and check $\delta g_L^b/\Delta T$ against CPSW Fig. 1.
3. **Custodial $g_R^b=0$ rep not pinned.** Sourced only for $b_R \in (1,1)_{-1/3}$ (ACDP Eq. 22 / Table 1), where $T_L^3=T_R^3=0 \Rightarrow \delta g_R^b=0$ by P_C . The code hard-codes the protected branch (`bR_strategy='elementary_zero'`) without recording the embedding that earns it. *To close:* declare $b_R \in (1,1)_{-1/3}$ as the production embedding and cite ACDP Table 1 row 2 (*closable now*).
4. **Singlet-only ΔT truncation.** The singlet ΔT (Eq. 13) is exact, but the dominant, *negative* bidoublet ΔT is omitted unless the override is set. *To close:* reconstruct the bidoublet ΔT , or confirm the override path is exercised in production.

What the full custodial build (roadmap D) would add. Per the literature catalog, a custodial implementation should reproduce: (i) the S -limited $\sim 2\text{--}3$ TeV EW floor; (ii) the $-T$ bidoublet / $+T$ singlet loop competition (CPSW Eqs. 41–43); (iii) the *unchanged* ε_K wall (Blanke); and (iv) the b_R -embedding-dependent A_{FB}^b . Reproducing Blanke Fig. 9 (custodial ε_K vs. M_{KK}) inside our own custodial code would be the headline validation. None of these change the note’s thesis — they sharpen the custodial *phenomenology* the proxy deliberately defers.

6 Conclusion

Constraint by constraint, custodial symmetry does exactly two things to what we run: it protects the oblique T parameter (pulling the S, T, U floor from our minimal $\sim 16\text{--}20$ TeV down to ~ 6 TeV, with the residual set by the unprotected S) and it protects the $Z b_L \bar{b}_L$ vertex at tree level (with top-partner loops partially undoing it). It does **nothing** to ε_K or any other flavor constraint: the

binding kaon operator is the color-octet KK-gluon C_4 , an electroweak singlet, so the wall stays at tens of TeV. This is not a limitation of our proxy — Blanke’s *full* custodial $\Delta F = 2$ calculation finds the same 18–30 TeV wall. **Custodial is an EW cure, not a flavor cure.** That conclusion is adequate for the note’s thesis; it is not the last word on custodial phenomenology, which the four deferred proxy items (§5) would address. Only *flavor alignment* (RH-down U(3) degeneracy / 5D-MFV / V_{5KM}) moves the ε_K wall.